

STM32H743 Drone Controller Board

This board was designed including component selection, schematic capture and PCB layout during one weekend.

The board is based on the **STM32H743VIT6** and contains a DC/DC power supply based on the **LT8609SEV#PBF ultra-low-EMI SMPS**, with optimized component placement, short high-current paths and a minimized switching hot loop to improve EMC performance, reduce parasitic inductance and maintain high power efficiency.

It also contains a **W25Q512JVEIQ Flash memory** with short high-speed routing, and a **Bosch BMI088 IMU** for inertial sensing using an integrated accelerometer and gyroscope.

The board uses a **4-layer stackup**, where:

- 1 – TOP:** signals, components, local GND copper and power routing
- 2 – GND PLANE:** solid ground plane providing low-inductance return current paths, good signal reference and improved EMC performance
- 3 – POWER PLANE:** low-impedance power distribution plane
- 4 – BOTTOM:** secondary signal routing mixed with GND copper